Mamadou Ben Hamidou CISSOKO Ph.D.

AI Research Scientist · Deep Learning, Time Series, LLMs & Multi-Agent Systems

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PROFILE

Ph.D. in Computer Science and postdoctoral AI researcher specialising in deep learning for irregular time series, large language models (LLMs), retrieval-augmented generation and multi-agent systems. I design recent neural architectures (JEPA, diffusion, state-space models) and take them from research to working systems. Experienced in leading interdisciplinary research with clinicians, teaching at Master's and Bachelor's level, and disseminating results (4 peer-reviewed publications; reviewer for NeurIPS, UAI and AISTATS).

PROFESSIONAL EXPERIENCE

LS2N — Nantes Digital Sciences Laboratory (UMR 6004)
Nov. 2024 – present

Postdoctoral Researcher · Digital twin and patient simulation under general anaesthesia — *Nantes, France*

- Lead the technical architecture of a high-fidelity physiological patient simulator for clinical decision support and medical training.
- Designed a digital-twin engine based on dynamic temporal neural networks that simulates multivariate intraoperative physiological changes in real time.
- Built distributed AI streaming pipelines for low-latency inference, real-time personalisation and model interpretability under clinical constraints.

Stack: PyTorch, TensorFlow, Python, Unity integration, distributed time-series frameworks, HPC

ICube — Engineering, Computer Science and Imaging Laboratory (UMR 7357)
Oct. 2020 – Oct. 2024

Ph.D. Researcher · Deep temporal networks for irregular ICU trajectories — *Strasbourg, France*

- Designed and open-sourced two dynamic-memory architectures, MWTA-LSTM and AMITA-LSTM, for sparse and irregular electronic health records.
- Formulated variable-level attention and frequency weighting that give direct interpretability, validated with ICU department heads.
- Built automated ingestion and preprocessing pipelines for large clinical databases (including MIMIC), using adaptive pooling to reduce distortions from extreme values.

Stack: PyTorch, TensorFlow, Python, scikit-learn, pandas, Git, HPC (Slurm)

ICube — University of Strasbourg
Apr. 2020 – Sep. 2020

Master's Research Intern · Deep learning for interpretability and anomaly detection — *Strasbourg, France*

- Developed a hybrid sequential network combining LSTM layers with automated shapelet extraction to detect temporal anomalies while keeping the model transparent.
- Applied class activation maps (CAM) and explainable-AI methods to visualise decision boundaries on health-state sequences.

Stack: Keras, TensorFlow, Python, shapelets

IFI — Francophone International Institute, Vietnam National University
Oct. 2018 – May 2019

Supervised research project · Machine learning for econometrics — *Hanoi, Vietnam*

- Compared classical econometric models with neural networks (ANN/RNN) for capturing nonlinear temporal dependencies in economic series.

Stack: Python, Keras, TensorFlow, R, statistical modelling

RESEARCH & ENGINEERING PROJECTS

TF-JEPA, FASTER & DyFAIP — joint imputation and prediction
2025 – 2026

JEPA, diffusion, state transitions, irregular time series

- **TF-JEPA**: time-frequency joint-embedding predictive architecture for structure-guided imputation of missing values.
- **FASTER & DyFAIP**: frequency-conditioned diffusion and state-transition models for joint imputation and trajectory prediction on highly sparse clinical timelines.

Drone micro-target detection — RT-DETRv2 with a DINOv2 backbone
2025 – 2026

RT-DETRv2, DINOv2 ViT-B, SAHI/ASAHI, false-negative audit

- Integrated a self-supervised DINOv2 ViT-Base backbone (768-dim) with multi-scale projection (P3–P5) into RT-DETRv2 to preserve representations of targets only a few pixels wide.
- Two-pass inference on 20 MP images: a global 3200 $\times$ 3200 scan followed by dynamic SAHI tiling with Greedy NMM merging.
- Post-validation audit isolating misses by IoU, with confusion matrices adapted to rare-class oversampling.

Drone small-object detection with Top-K spatial attention 2025 – 2026

YOLOv8 / YOLO26-P2, Top-K attention, DBSCAN, Grad-CAM, CLAHE

- Designed a dynamic spatial-attention projector (ActiveTopKFeatureProjector) that re-injects high-resolution P2/P3 activation maps into the detection head.
- Multi-process tiling guided by DBSCAN clustering, CLAHE contrast enhancement and adaptive oversampling of rare species.
- Multi-scale SAHI/ASAHI sliced inference and multi-level Grad-CAM to track saliency retention through deep layers.

Multivariate time-series anomaly detection 2025 – 2026

MTGNN, GNN-LSTM, SWaT, WADI, TensorBoard

- Implemented and evaluated graph-based detectors (MTGNN, GNN-LSTM) on multivariate industrial data (SWaT, WADI), with a complete training, evaluation and visualisation framework.

Conversational commerce agent — hybrid RAG and tool execution 2026

BM25 + dense, RRF, cross-encoder, Pydantic, DeepEval, Gradio

- RAG pipeline combining dense (SentenceTransformers) and sparse (BM25) retrieval, fused with Reciprocal Rank Fusion and reranked by a cross-encoder (ms-marco-MiniLM-L-6-v2).
- Agent that routes each request to the document base or to tools (order tracking, cart management with Pydantic).
- Automated PyTest/DeepEval quality gates (faithfulness, contextual relevance) and a Gradio dashboard for inspecting traces, reranker scores and tool arguments.

Multi-agent systems and tool-calling frameworks 2026

LangGraph, React, Vite, Model Context Protocol (MCP), CI/CD

- Modular agent pipelines for web research, code review, PDF question answering and multi-agent debate.
- Full-stack applications with a React/Vite front end, MCP server integrations and automated GitHub CI/CD.

Agent state orchestration and hybrid RAG 2026

LangGraph, state management, dense and sparse retrieval

- LangGraph agent workflows with advanced state and multi-step execution-path management, plus dense/sparse hybrid RAG for precise grounding.

LLM observability and monitoring pipelines 2026

OpenTelemetry, Prometheus, tracing, latency

- Tracing and monitoring architecture with OpenTelemetry and Prometheus, tracking agent execution paths, token latency and tool-call reliability under load.

Deep NLP and generative recommender systems 2018 – 2020

Seq2Seq, BiLSTM, transfer learning, RBM

- PyTorch recommender based on restricted Boltzmann machines, and a multi-turn conversational agent (attention Seq2Seq, BiLSTM, domain transfer learning).

PUBLICATIONS

J = journal · C = conference · S = submitted

[J.1] **Cissoko, M. B. H.**, Castelain, V., et al. (2025). Predicting and Interpreting Healthcare Trajectories from Irregularly Collected Sequential Patient Data Using AMITA. *Information Sciences*, 705. **[Link]**

[J.2] **Cissoko, M. B. H.**, Castelain, V., et al. (2025). Multi-Way Adaptive Time-Aware LSTM for Irregularly Collected Sequential ICU Data. *Expert Systems with Applications*, 261. **[Link]**

[C.1] **Cissoko, M. B. H.**, et al. (2024). Modeling Temporal Dynamics in Irregular ICU Data Using MWTA-LSTM. *EPIA 2024*, Springer LNCS. **[Link]**

[C.2] **Cissoko, M. B. H.**, et al. (2024). Predicting Patient Health Outcomes with AMITA from Irregular Time Series. *ICONIP 2024*, Springer LNCS. **[Link]**

[S.1] **Cissoko, M. B. H.**, Sinoquet, C., et al. TF-JEPA: Time-Frequency Joint Embedding Predictive Architecture for Incomplete Time Series. *Under review.*

[S.2] **Cissoko, M. B. H.**, Sinoquet, C., et al. DyFAIP — End-to-End Joint Imputation and Prediction in Irregular Time Series via Adaptive Frequency Transitions. *Expert Systems with Applications* (under review).

[S.3] **Cissoko, M. B. H.**, Sinoquet, C., et al. FASTER: Frequency-Conditioned Diffusion for Joint Imputation and Prediction in Sparse Time Series. *Expert Systems with Applications* (under review).

TECHNICAL SKILLS

Deep learning & R&D	PyTorch (DDP, FSDP), TensorFlow/Keras, JEPA, diffusion models, SSM/Mamba, LSTM/GRU, Transformers, GNNs, XAI (CAM, Grad-CAM)
LLMs & generative AI	Hybrid RAG, LangGraph, MCP, fine-tuning, prompt engineering, Hugging Face, vLLM, DeepEval
Computer vision	RT-DETRv2, DINOv2, YOLO, SAHI/ASAHI, small-object detection
Optimisation & inference	Quantisation (INT8/FP16), distillation, pruning, ONNX Runtime, TensorRT, Triton Server
MLOps & infrastructure	Docker, Git/GitHub, CI/CD, PyTest, MLflow, W&B, OpenTelemetry, Prometheus, HPC (Slurm)
Data & health informatics	scikit-learn, pandas, NumPy, SciPy, EHR / MIMIC-III, FHIR, HL7
Languages	Python, C++, Java, R, SQL, SPARQL, RDF/XML, JSON-LD

TEACHING

Nantes Université — Nov. 2024 – present

Lecturer — Master's and Bachelor's programmes

- **Master's**: sequential architectures (LSTM, GRU), Transformers, optimisation and temporal feature engineering.
- **Bachelor's**: statistical inference, supervised and unsupervised learning (SVM, random forests), evaluation metrics.
- Practical sessions in PyTorch and scikit-learn; supervised 60+ engineering students.

Vietnam National University — Dec. 2024 – present

Lecturer — Master's programme

- Sequential modelling, attention mechanisms and temporal data; code reviews and workshops on PyTorch, cross-validation, version control and reproducibility.

PEER REVIEW & ACADEMIC SERVICE

- Conferences: NeurIPS, UAI, AISTATS.
- Journals: *Information Sciences* (30+ reviews), *Environmental Technology & Innovation* (2), *Neurocomputing* (1).

EDUCATION

Ph.D. in Computer Science, Mathematics and Engineering — 2020 – 2024
University of Strasbourg (ICube)

M.Sc. Computer Science — Intelligent Systems and Multimedia — 2020
Vietnam National University (IFI), Hanoi

M.Sc. Computer Science — Data Engineering and Business Intelligence — 2020
University of La Rochelle

B.Sc. Computer Science Applied to Business Management — 2013
IUG, Institut Universitaire de Gestion, Bamako

AWARDS & CERTIFICATIONS

Finalist (Top 5) — Vietnam AI Grand Challenge	2019
TIBCO-IFI Excellence Scholarship	2020
CCNA (Cisco) and MCITP (Microsoft) certifications	2017

LANGUAGES

French (native) · English (fluent) · Vietnamese (basic)

REFERENCES

Prof. Nicolas Lachiche
Professor, University of Strasbourg; deputy head of the SDC team, ICube — Ph.D. supervisor
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Prof. Vincent Castelain
Professor and hospital practitioner, University of Strasbourg; head of the ICU, Hautepierre Hospital — Ph.D. supervisor
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Dr. Christine Sinoquet
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